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Japan Credit Strategy | Asia Pacific

Japan Corp Credit Primer –
Eight Facts You Need to Know

With renewed interest in Japan's corporate bond market, we compile eight facts that we believe interested investors need to know. From these facts, we also tease out key lessons and observations when investing in the asset class.

Fact #1: Japan's corp bond market is the second largest in APAC and growing: We think that it would be difficult for investors to ignore Japan's corp bond market given its size of ~US\$1.5 trillion ([Exhibit 1](#)), especially if it continues to grow at the current pace.

Fact #2: For Asia credit investors, Japan credit is a key source of diversification and credit risk: Japan credit can help to reduce economy concentration risk of the current Asia USD credit benchmark. At the same time, negative China credit net supply has turned Japan credit into an important source of credit risk for Asia credit investors.

Fact #3: US credit investors have exposure to Japan credit, and it has been increasing: After US credit, Japan credit is firmly the third-largest country exposure for the US IG index ([Exhibit 2](#)). In the past 10 years, aside from US credit, Japan credit's market share in the US IG index has the highest increase among countries in the index.

Fact #4: Japan USD credit lacks sector diversification currently, but this could change: Lack of sector diversification is a key pushback from investors on Japan USD credit. However, we think that this will change, as we expect higher Japan non-financial USD corp bond net supply.

Fact #5: Japan's LC corp bond market has well-balanced sector and duration distribution: We believe Japan's LC corp bond market's good sector diversification shows the potential for Japan USD credit to improve its sector diversification.

Fact #6: Large Japanese corporates are not averse to debt funding: We expect more Japanese corporates to tap the FC corp bond market as it is a natural progression after being comfortable with tapping the LC corp bond market.

Fact #7: Demand for Japan USD credit from investors has increased in the past two years: Geographically, US and Asia credit investors have been increasing their allocation towards Japan USD credit in the past two years. In terms of investor type, banks saw the largest increase in demand for Japan USD credit.

Fact #8: The Japan IG spread looks cheap versus global peers, especially on a spread/duration basis: Despite its tight spread, we see Japan IG having the most room to tighten among global IG peers because it is trading wide versus its historical tights.

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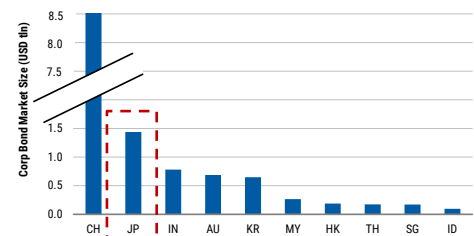
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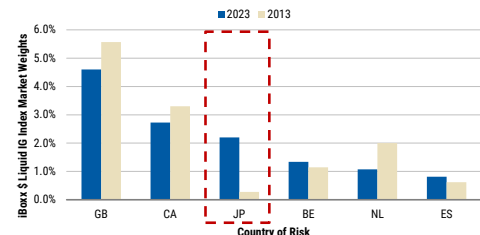
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Exhibit 1 : Japan's corp bond market is currently the second largest in APAC at ~US \$1.5 trillion



Source: Bloomberg, Morgan Stanley Research; Note: Data as of December 2023.

Exhibit 2 : Japan is the third-largest exposure aside from US credits for the iBoxx \$ Liquid IG Index



Source: iBoxx, Morgan Stanley Research

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Executive Summary

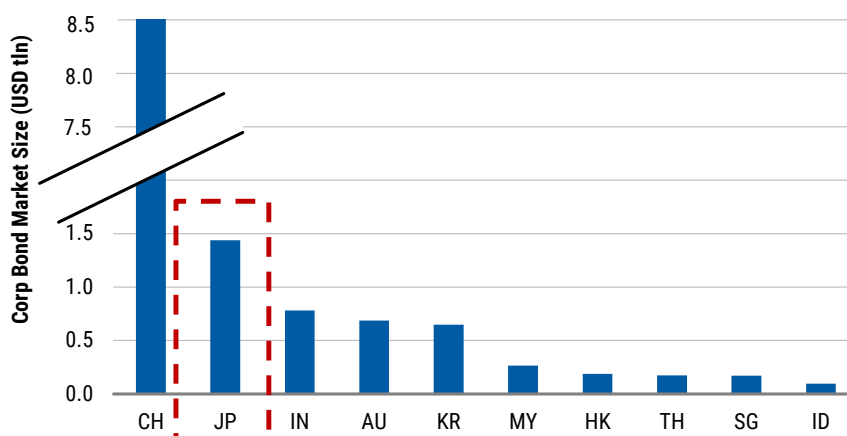
With renewed interest in Japan's corp bond market, we compiled eight key facts that we believe interested investors would need to know. Below is a list of all eight facts and links to key sections of this report:

- **Fact #1: Japan's Corp Bond Market Is the Second Largest in APAC and Growing**
- **Fact #2: For Asia Credit Investors, Japan Credit Is a Key Source of Diversification and Credit Risk**
- **Fact #3: US Credit Investors Have Exposure to Japan Credit, and it Has Been Increasing**
- **Fact #4: Japan USD Credit Lacks Sector Diversification Currently, but This Could Change**
- **Fact #5: Japan's LC Corp Bond Market Has Well-Balanced Sector and Duration Distribution**
- **Fact #6: Large Japanese Corporates Are Not Averse to Debt Funding**
- **Fact #7: Demand for Japan USD Credit from Investors Has Increased in the Past Two Years**
- **Fact #8: The Japan IG Spread Looks Cheap versus Global Peers, Especially on a Spread/Duration Basis**

That said, not all investors will have the time to go through all eight facts in detail. Hence, we try our best below to tease out the key lessons and observations investors should come away with when going through the facts:

Fact #1: Japan's Corp Bond Market Is the Second Largest in APAC and Growing: We think that it would be difficult for investors to ignore Japan's corp bond market given its size, especially if it continues to grow at the current pace. We identify three growth drivers for the LC corp bond market and see M&A as an important driver of FC corp bond market growth.

Exhibit 3: Japan's corp bond market is currently the second largest in APAC at ~US \$1.5 trillion

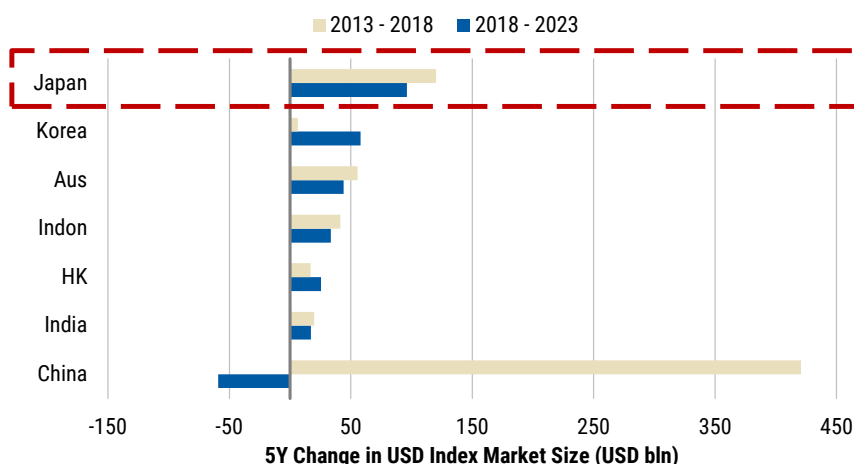


Source: Bloomberg, Morgan Stanley Research; Note: Data as of December 2023.

Fact #2: For Asia Credit Investors, Japan Credit Is a Key Source of Diversification and

Credit Risk: Japan credit can help to reduce economy concentration risk of the current Asia USD credit benchmark. We believe that this could be the most natural diversification path for Asia credit investors. With the decline in China credit net supply, Japan credit has been an important source of credit risk for Asia credit investors.

Exhibit 4: Japan credit's steady market size growth is overshadowed by China credit's more volatile growth

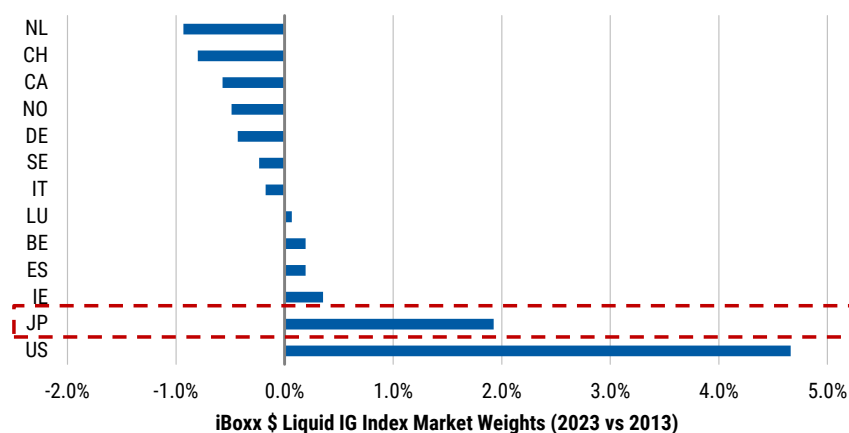


Source: iBoxx, Morgan Stanley Research

Fact #3: US Credit Investors Have Exposure to Japan Credit, and it Has Been Increasing:

After US credit, Japan credit is firmly the third-largest country exposure for all three US IG indices. In the past 10 years, aside from US credit, Japan credit's market share in all three US credit IG indices has seen the largest increase among countries in the indices.

Exhibit 5: Japan credit's market share in the iBoxx \$ Liquid IG index saw the largest increase, outside of US credit



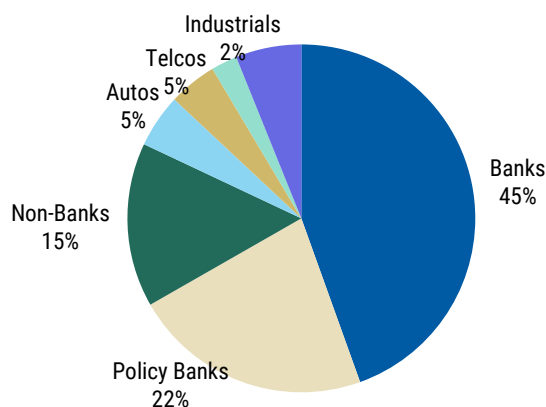
Source: iBoxx, Morgan Stanley Research

Fact #4: Japan USD Credit Lacks Sector Diversification Currently, but This Could

Change: Japan USD credit's key drawback is that it lacks sector diversification. Financials is currently 82% of Japan USD credit; the top three sectors are all financials – banks, policy banks and non-banks. However, we think that this will change, as we expect higher Japan non-financial USD corp bond net supply from overseas M&A (Fact #1).

Exhibit 6: The financials sector is 82% of Japan USD credit sector exposure

Japan USD Credit Sector Breakdown (USD 334 bln)



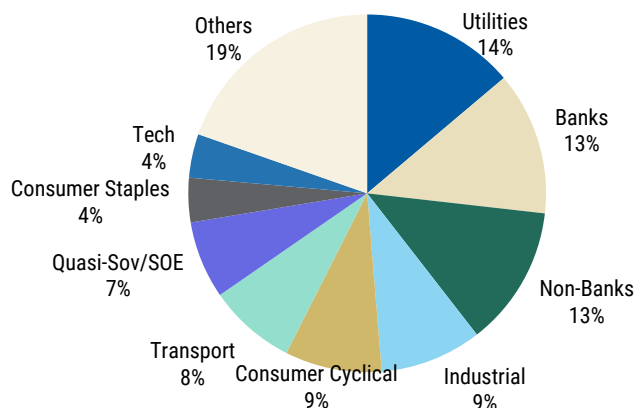
Source: iBoxx, Morgan Stanley Research

Fact #5: Japan's LC Corp Bond Market Has Well-Balanced Sector and Duration

Distribution: Japan's LC corp bond market has a balanced maturity profile in contrast to the maturity profile of most APAC LC corp bond markets. We believe that Japan's LC corp bond market's good sector diversification shows the potential for Japan USD credit to improve its sector diversification, which is supportive of [Fact #4](#).

Exhibit 7: No single sector in Japan's LC corp bond market has more than a 15% market share

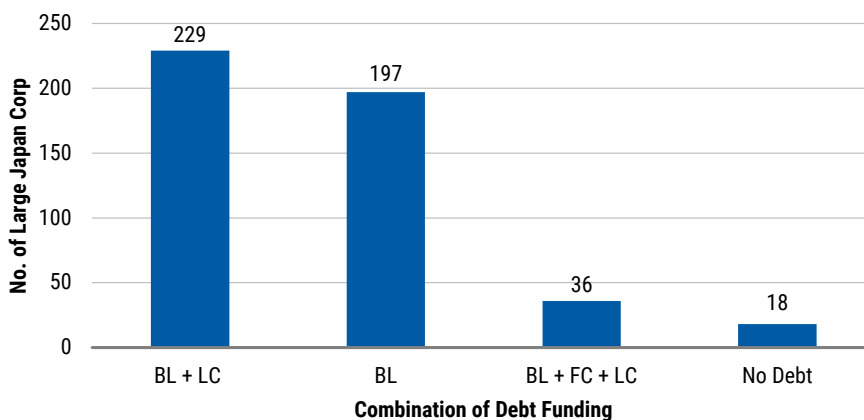
Japan LC Corp Bond Market Sector Breakdown (USD 655 bln)



Source: Bloomberg, Morgan Stanley Research

Fact #6: Large Japanese Corporates Are Not Averse to Debt Funding: Large Japanese corporates have been using bank loans, LC corp bonds and FC corp bonds to fund their debt. We expect more Japanese corporates to tap the FC corp bond market as it is a natural progression after being comfortable with tapping the LC corp bond market. It is currently cheaper for Japan IG corporates to issue in the USD bond market and hedge it than to issue LC corp bonds.

Exhibit 8: Most large Japanese corporates have some combination of debt on their balance sheet

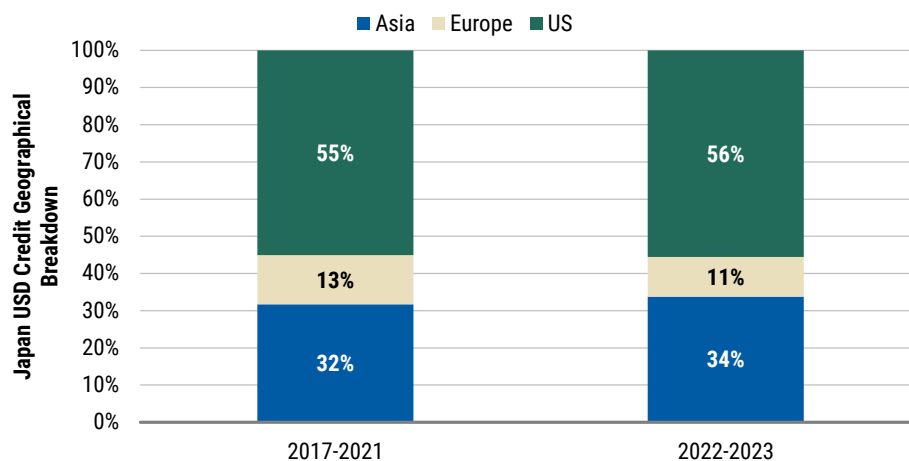


Source: Bloomberg, Morgan Stanley Research; Note: BL = bank loans, FC = FC corp bonds, LC = LC corp bonds.

Fact #7: Demand for Japan USD Credit from Investors Has Increased in the Past Two Years:

Both US and Asia credit investors had been increasing their allocation towards Japan USD credit in the past two years. Banks have seen the largest increase in allocation towards Japan USD credit in the past two years due to the lack of high-quality Asia credit net supply, especially in the financial sector.

Exhibit 9: Asia and US credit investors have increased their allocation to Japan USD credit in the past two years



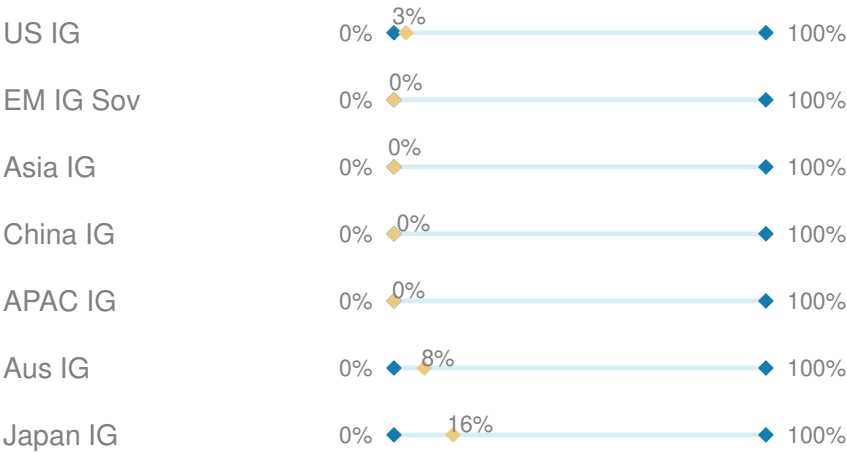
Source: Bloomberg, Morgan Stanley Research

Fact #8: The Japan IG Spread Looks Cheap versus Global Peers, Especially on a Spread/Duration Basis:

Despite its tight spread, we see Japan IG having the most room to tighten among global IG peers because it is trading at the 16th percentile of its historical spread. Japan IG has the shortest duration among global IG peers and looks cheap versus its global peers on a spread/duration basis, thanks to the global IG long-end spread being rich.

Exhibit 10: From a spread percentile rank, Japan IG is currently the cheapest global IG credit

Spread Ranges - Percentile Rank (Current vs. History)



Source: iBoxx, Bloomberg, Morgan Stanley Research; Note: All IG credit history starts from January 1, 2013.

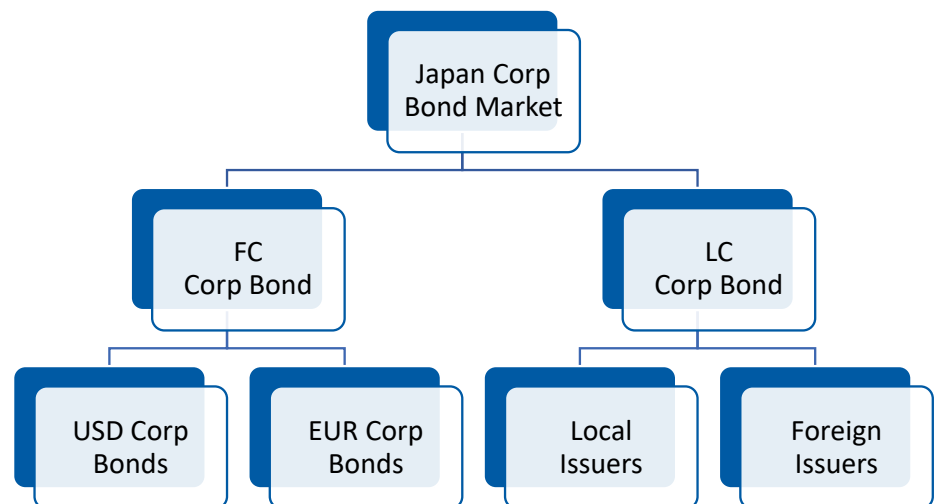
Defining Japan's Corp Bond Market

Before we start listing the eight facts we believe investors need to know about Japan's corp bond market, we think it is important to first define its scope to avoid any room for ambiguity. When we look at Japan's corp bond market, we think that it would make sense to include both the foreign currency (FC) and local currency (LC) corp bond markets ([Exhibit 11](#)).

A typical Japanese corporate has the option to raise its corp bonds in either the LC (JPY in this case) or the FC corp bond market. In contrast, US and European corporates don't have this ability, and mainly fund their corp bonds in either USD or EUR. This distinctive feature is not only for Japan corporates, but also for most corporates in the Asia Pacific (APAC) region such as China and India. Hence, if we want to take a holistic approach to analysing Japan's corp bond market, we would need to include both the LC and FC corp bond markets.

We can divide Japan's LC corp bond market into corp bonds from Japanese issuers and those from foreign issuers. LC corp bonds from foreign issuers are known as Samurai bonds. In theory, Japan's FC corp bond market could include any currency aside from JPY, but the two most used currencies by Japanese corporates are USD and EUR.

Exhibit 11: Breaking down Japan's corp bond market



Source: Morgan Stanley Research

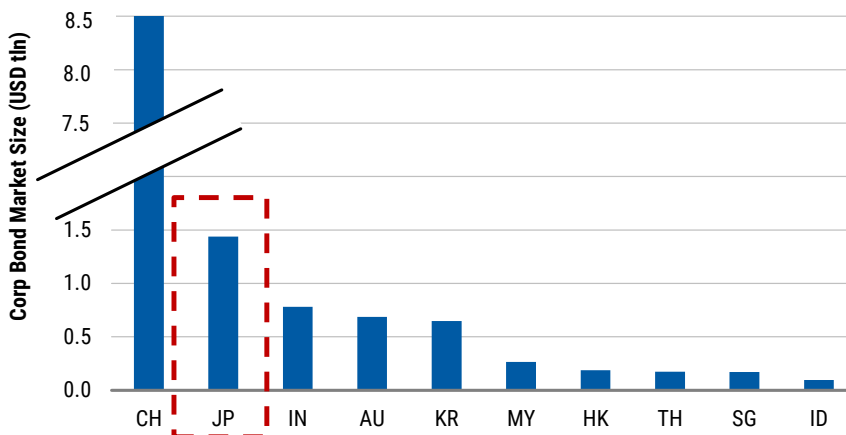
Fact #1: Japan's Corp Bond Market Is the Second Largest in APAC and Growing

- Japan's corp bond market is sizeable from a global and APAC region perspective. We think that it would be difficult for investors to ignore how large it is, especially if it continues to grow at the current pace.
- Japan LC corp bonds experienced a paradigm shift towards greater market size growth in the past four years, driven by: 1) Strong Japan corporate capex; 2) Tightening monetary policy by the Bank of Japan (BoJ); and 3) Corporate reforms.
- Japan's overseas M&A deals have been an important driver of higher Japan FC corp bonds' net supply. We see upside potential to Japan FC corp bonds' net supply this year, given that we expect more Japan overseas M&A deals going forward.

After China, Japan has the second-largest corp bond market in the APAC region

Although Japan's nominal growth has been range-bound for most of the past decade, its corp bond market size is the second largest in APAC ([Exhibit 12](#)). At the end of 2013, Japan's corp bond market was close to US\$1.1 trillion in size, just US\$800 billion behind China. Ten years later, Japan's corp bond market has grown by another US\$350 billion to US\$1.5 trillion.

Exhibit 12: Japan's corp bond market is currently the second largest in APAC at ~US \$1.5 trillion

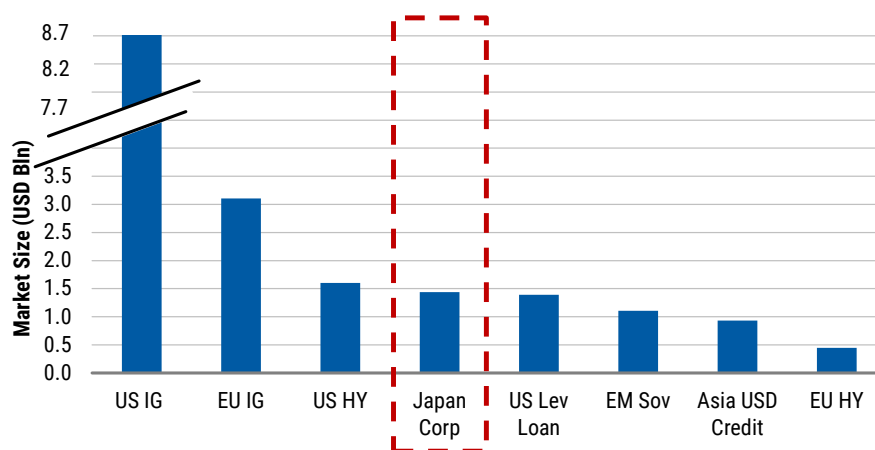


Source: Bloomberg, Morgan Stanley Research; Note: Data as of December 2023.

Although it remains the second-largest corp bond market in the APAC region, its size is significantly dwarfed by China's corp bond market. At US\$8.5 trillion, China's corp bond market is currently the largest in APAC by a wide margin. That said, excluding China, Japan's corp bond market is much larger compared to the rest of the region; it is 1.8x the size of India's corp bond market, which is the third largest.

From a global credit perspective, the size of Japan's corp bond market is between that of the US HY and US leveraged loan markets. It is currently ranked fourth when compared to the global credit asset class that most global credit investors can invest in. Japan's corp bond market is US\$300-500 billion higher than both the EM sovereign bond market and the Asia credit USD bond market.

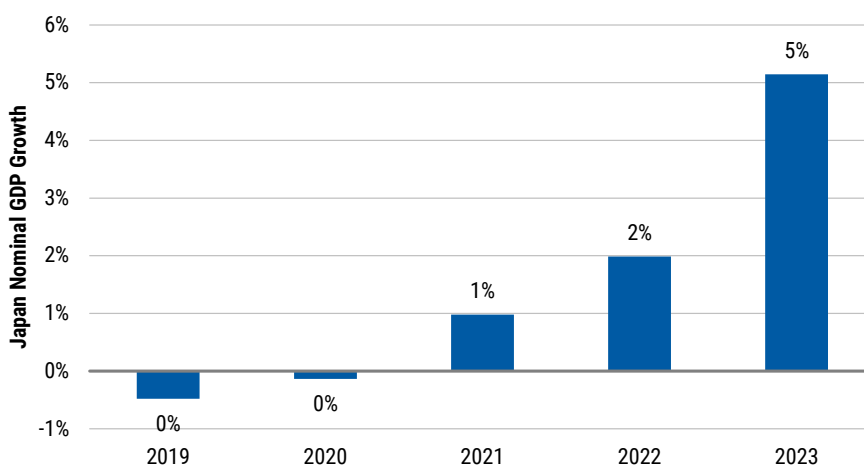
Exhibit 13: The size of Japan's corp bond market is between that of the US HY and the US leveraged loan markets



Source: Bloomberg, Morgan Stanley Research; Note: Data as of December 2023.

One reason behind Japan's corp bond market's second-placed position among other corp bond markets in the region is its strong growth in the past three years, which has been driven by a strong post-Covid recovery. The country's nominal GDP has increased post-Covid and reached a high of 5%Y growth last year (Exhibit 14).

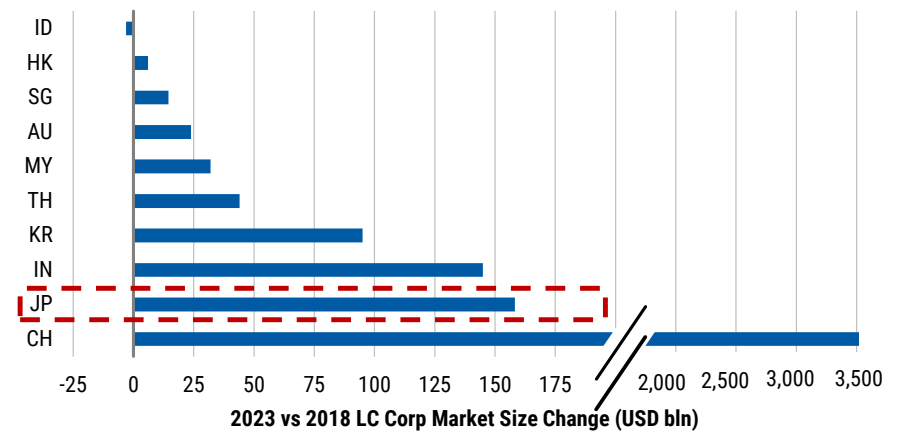
Exhibit 14: Post-Covid, Japan's nominal GDP growth has improved significantly



Source: CEIC, Morgan Stanley Research

Among APAC's LC corp bond markets, Japan's corp bond market has seen the second-highest growth in the past five years in nominal terms ([Exhibit 15](#)). Its US\$158 billion increase in the past five years is slightly higher than India's US\$145 billion increase, but much lower than China's US\$3.4 trillion increase.

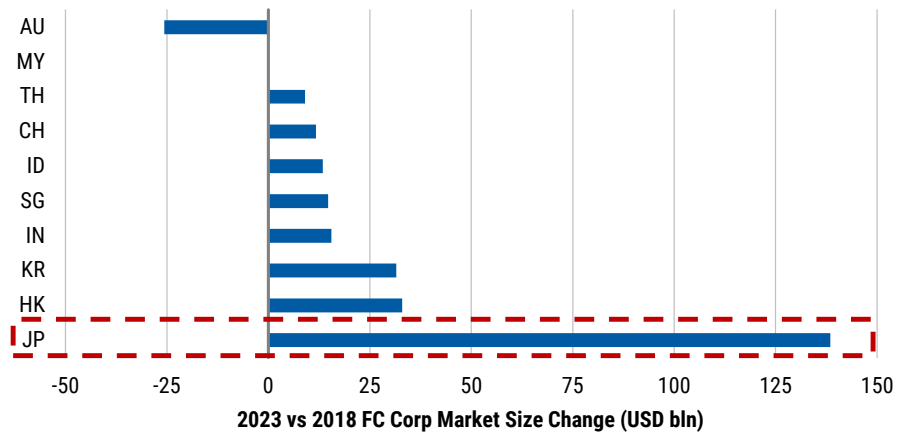
Exhibit 15: Japan's LC corp bond market has seen the second-largest increase in APAC after China



Source: Bloomberg, Morgan Stanley Research

When we look at the nominal growth of APAC's FC corp bond market in the past five years, Japan's corp bond market stands out as the market with the largest increase in size ([Exhibit 16](#)). Japan's FC corp bond market has increased by US\$138 billion in the past five years, 4-4.5x higher than Hong Kong (ranked second) and Korea (ranked third) corp bond markets' increase over the same period.

Exhibit 16: Japan has seen the largest increase in FC corp bond market size in the past five years among APAC economies



Source: Bloomberg, Morgan Stanley Research

Japan's LC corp bond market post-Covid growth is the highest since 2000

We break down the growth of Japan's LC corp bond market into three stages ([Exhibit 17](#)):

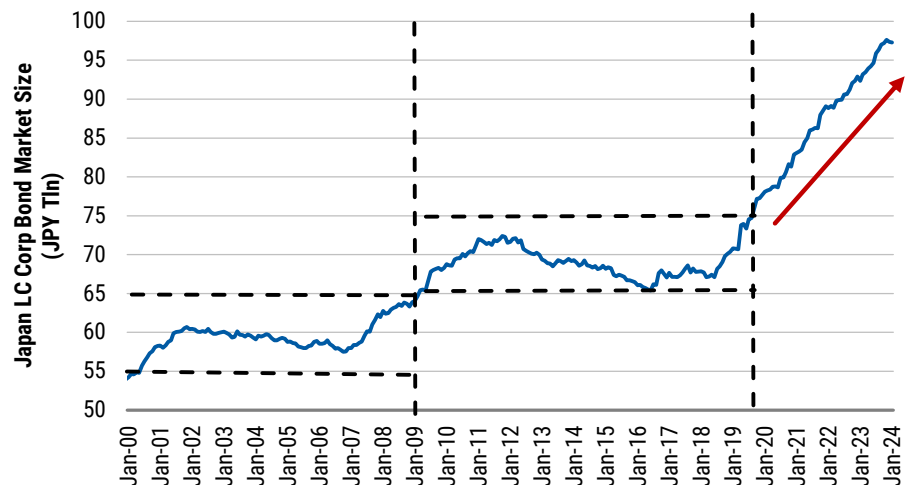
Stage #1: Pre-GFC (2000-09)

Stage #2: Post-GFC (2010-19)

Stage #3: Post-Covid (2020 to now)

In the first two stages of pre-GFC and post-GFC, Japan's LC corp bond market only managed to grow by around JPY 10 trillion in ten years, at an average 1.1.5%Y annual growth rate. However, in stage 3, it managed to increase by ~JPY 20 trillion in just three years, at an average annual growth rate of 8.4%Y. It is clear that stage 3's growth rate is much higher than in the other two stages.

Exhibit 17: Japan's LC corp bond market growth in the past four years has been unprecedented



Source: CEIC, Morgan Stanley Research

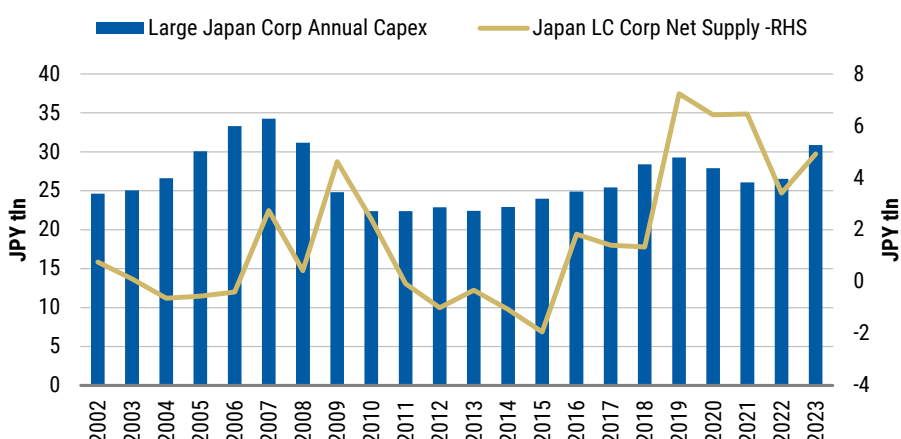
So what drove Japan's LC corp bond market's growth in the past three years? We see three key drivers that can help to explain stage 3's strong market size growth for Japan LC corp bonds:

- **Strong Japan corporate capex post-Covid:** In our view, capex tends to be the main driver of corp bonds' net supply. If we compare Japan's large corporate capex trend with its LC corp bonds' net supply, we find that higher capex tends to drive LC corp bonds' net supply higher and vice versa ([Exhibit 18](#)). More importantly, the post-Covid capex level of large Japan corporates trended higher than during the post-GFC period, which drove the strong pick-up in Japan LC corp bonds' net supply in the past four years.
- **Tightening monetary policy by the Bank of Japan (BoJ):** Even before the removal of the negative interest rate policy (NIRP) and yield curve control (YCC) in March this year (see [Farewell, NIRP and YCC as Expected: Quick Impression Before Gov. Ueda's Press Conference](#), March 19, 2024), most Japan corporates in 2023 had expected the BoJ to start tightening its monetary policy in the next 6-12 months, which means that the cost of LC corp bond debt would increase. Hence, we

suspect that some Japan corporates might have front-loaded some of their LC corp bond issuance in 2023 to lock in a lower funding cost.

- **Japanese corporate reforms:** As pointed out by our Japan equity strategy team, corporate reform remains a key focus for Japanese corporates (see [Catching the Corporate Reform Wave - Key Opportunities in Japan's ROE Journey](#), Jun 15, 2023) due to the Tokyo Stock Exchange (TSE)'s calls for corporates with a low price-to-book ratio to improve capital efficiency and return on equity (ROE). We believe that a greater focus on capital efficiency would encourage Japan corporates to raise more corp bonds due to their low cost of funding and tax benefit. That said, we do think that we are in early stages of this corporate reform cycle and it will take some time before we can see any big shift in using LC corp bonds as a way to increase capital efficiency.

Exhibit 18: Capex is an important driver of Japan LC corp bonds' net supply



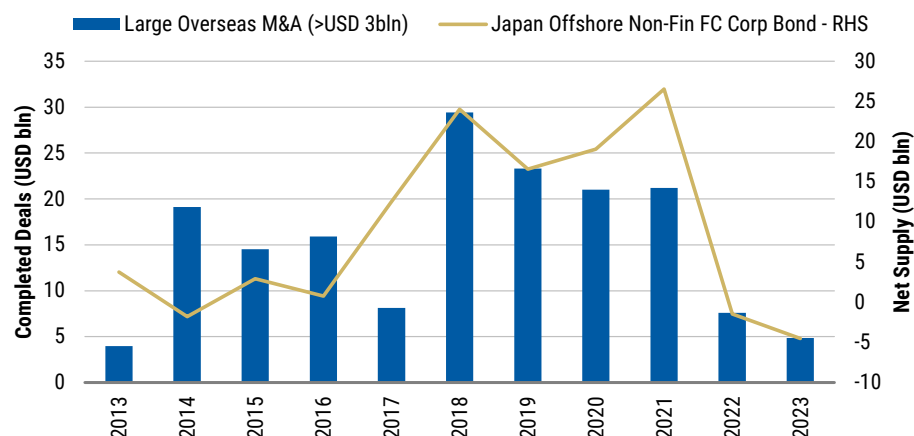
Source: CEIC, Morgan Stanley Research

Overseas M&A is supportive of higher Japan USD corp bond market growth

One of the more obvious reasons why a Japan corporate taps the FC corp bond market is because it needs to fund a large overseas M&A deal. For example, if a Japanese corporate acquires a company in the US, it would make sense to fund the acquisition with USD-denominated corp bonds to reduce any currency mismatch. In addition, the FC corp bond market allows a Japanese corporate to issue corp bonds with a long duration (>5 years), which would help Japanese corporates to manage refinancing risk.

However, the size of the overseas M&A deal would need to be large enough for Japanese corporates to tap the FC corp bond market. Otherwise, they could easily fund overseas M&A through bank loans or internal cash. We would think that a large overseas M&A deal, with a size of at least US\$3 billion, would be sufficient to incentivise a Japanese corporate to tap the FC corp bond market. Between 2018 and 2021 we saw a strong pick-up in completed overseas M&A deals that were over US\$3 billion, which coincided with the strong increase in net supply of non-financial FC corp bonds ([Exhibit 19](#)). Most of the large overseas M&A deals during this period were funded partially by FC corp bonds, mainly in either USD or EUR.

Exhibit 19: Large Japanese corporates' overseas M&A deals tend to drive Japan USD corp bonds' net supply higher



Source: Bloomberg, Morgan Stanley Research; Note: 2018 completed deal excludes Takeda Pharmaceuticals' takeover of Shire, given its outlier takeover size of US\$80 billion; Large M&A deals exclude Japanese financials' overseas M&A deals.

The largest overseas M&A during this period was Takeda Pharmaceuticals' takeover of Shire for US\$80 billion in 2018. Takeda Pharmaceuticals issued US\$5.5 billion of USD-denominated corp bonds and €7.5 billion of EUR-denominated corp bonds to fund the acquisition. Another example would be Seven & i Holdings Co Ltd's US\$21 billion acquisition of Speedway LLC in 2020. Seven & i Holdings Co Ltd ended up issuing US\$10.9 billion of USD corp bonds across maturities to help to fund the acquisition. Naturally, we believe that the larger the M&A deal, the more likely Japanese corporates will issue USD or EUR-denominated corp bonds to help to fund the acquisition.

There are currently four large (>US\$3 billion) overseas M&A deals in Japan that are pending ([Exhibit 20](#)), which will potentially drive the Japan FC corp bonds' net supply higher this year, assuming these deals are completed.

Exhibit 20: There are currently four large overseas M&A deals in Japan that are pending

Deal Type	Announce Date	Target Name	Acquirer Name	Announced Total Value (USD Bln)	Deal Status
M&A	3/12/2024	Innomotics/Siemens AG	Nidec Corp,KPS Capital Partners LP	3.28	Proposed
M&A	2/15/2024	Altium Ltd	Renesas Electronics Corp	5.69	Pending
M&A	1/18/2024	MDC Holdings Inc	Sekisui House Ltd	4.41	Pending
M&A	12/18/2023	United States Steel Corp	Nippon Steel Corp	13.40	Pending

Source: Bloomberg, Morgan Stanley Research

For 2024, our Japanese equity strategy team expects more overseas M&A activity from Japanese corporates as they begin to expand their global reach (see [Japan: More Outbound, More Buyouts](#), March 4, 2024). Our Morgan Stanley analyst survey identifies seven sectors where our equity sector teams expect increased M&A activity over the next 12 months – Beverages, Pharmaceutical, Financials, Transportation, Retail, J-REITs and Logistics. Hence, given our expectation of more Japan overseas M&A deals going forward, we see upside potential to Japan's FC corp bond market's net supply this year.

Fact #2: For Asia Credit Investors, Japan Credit Is a Key Source of Diversification and Credit Risk

- Japan credit can help to reduce economy concentration risk of the current Asia USD credit benchmark. We believe that this could be the most natural diversification path for Asia credit investors.
- With the decline in China credit net supply, Japan credit has been an important source of credit risk for Asia credit investors.
- Global credit investors have exposure to Japan credit through US IG credit indices, and Japan credit is the third-largest country exposure in US IG credit indices (aside from US credits).

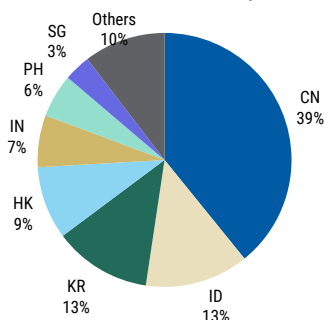
Japan credit reduces economy concentration risk in APAC credit

As we highlighted in [The Future of Asia Credit](#), September 8, 2023, we believe that Asia credit investors could boost their Asia credit portfolio's risk-adjusted return by broadening their investment horizon. In our view, the path of adding Japan credit into the current Asia credit benchmark is a natural choice for Asia credit investors.

By adding Japan credit into their current Asia credit portfolio, Asia credit investors can reduce their economy concentration risk significantly. If we look at the current Asia USD credit index, China exposure is 39% of the index, close to three times the exposure of the second-largest economy – Indonesia ([Exhibit 21](#)). Although China exposure is also the largest for the APAC USD credit index, it is only 8% higher than the second-largest economy in the index – Japan.

Exhibit 21: Compared to the Asia USD credit index...

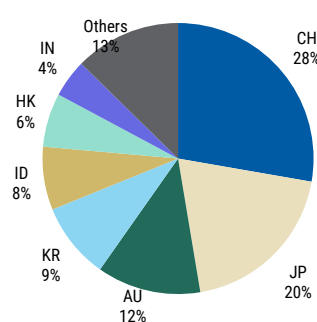
Bloomberg Asia USD Credit Index Economy Breakdown (Dec'23)



Source: Bloomberg, Morgan Stanley Research

Exhibit 22: ...the APAC USD credit index has much lower economy concentration risk

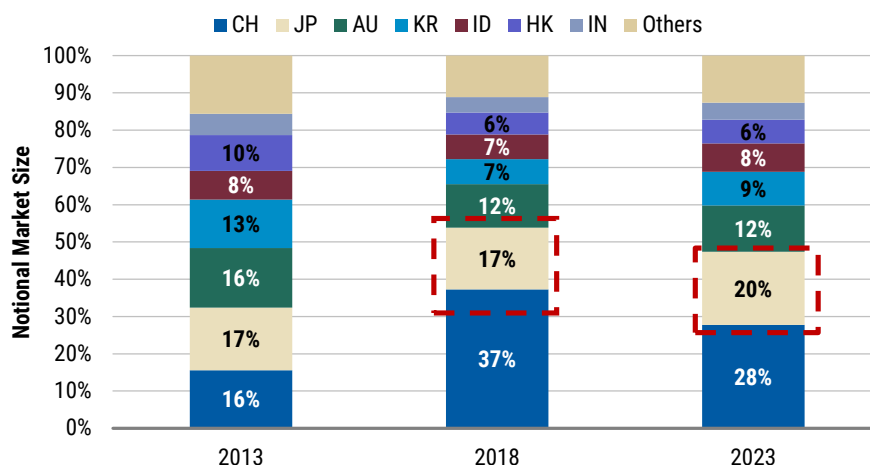
iBoxx APAC USD Credit Index Economy Breakdown (Dec'23)



Source: iBoxx, Morgan Stanley Research

Hence, Japan credit plays an important role in the APAC index by reducing economy concentration risk. It is the only economy within the APAC region with a market size that can counter-balance China credit's large market size. Since 2013, Japan credit remains the second-largest economy exposure for the APAC credit index ([Exhibit 23](#)). However, between 2018 and 2023, its market share saw the highest increase among APAC economies in the index, following the significant drop in China credit's market share.

Exhibit 23: Japan credit saw the largest market share gain within the iBoxx APAC credit index between 2018 and 2023

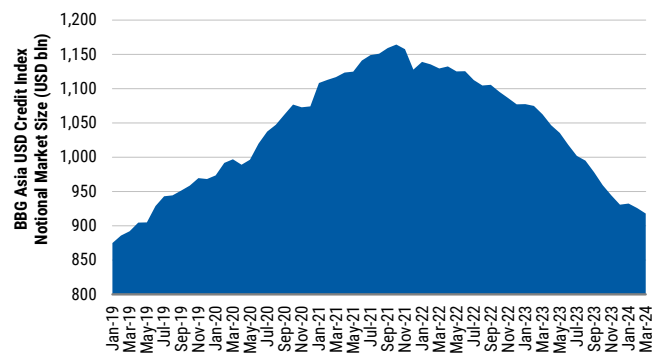


Source: iBoxx, Morgan Stanley Research

Japan credit has been a key source of credit risk for Asia credit investors

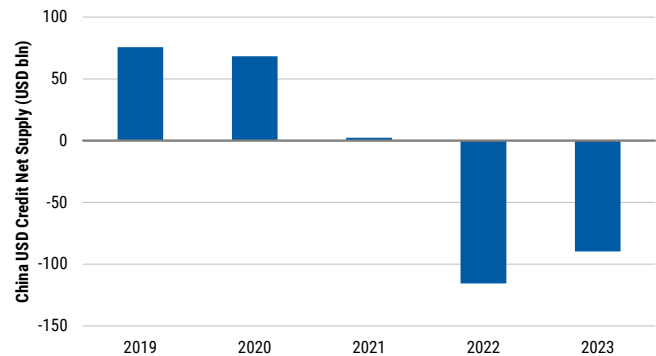
Aside from high economy concentration risk, another key concern for Asia credit investors is the shrinking market size of the Asia credit universe/index ([Exhibit 24](#)). Since the Asia credit market size peaked in late 2021, its market size has been shrinking for 23 months out of the past 29. This is mainly driven by China credit's large negative supply ([Exhibit 25](#)). Unfortunately, we don't expect China credit net supply to pick up this year; we are forecasting it to remain negative for the third year in a row (see [China Credit Net Supply To Remain Negative Next Year \(But Less So\)](#)..., November 12, 2023). Most Asia credit investors tend to agree with us, as shown in our recent global credit survey where 91% of Asia credit investors expect China credit net supply to be negative in 2024 (see [Asia Credit Investors Expect China Credit Net Supply To Remain Negative](#)..., March 8, 2024).

Exhibit 24: The Asia credit index market size has been shrinking since 2021...



Source: Bloomberg, Morgan Stanley Research

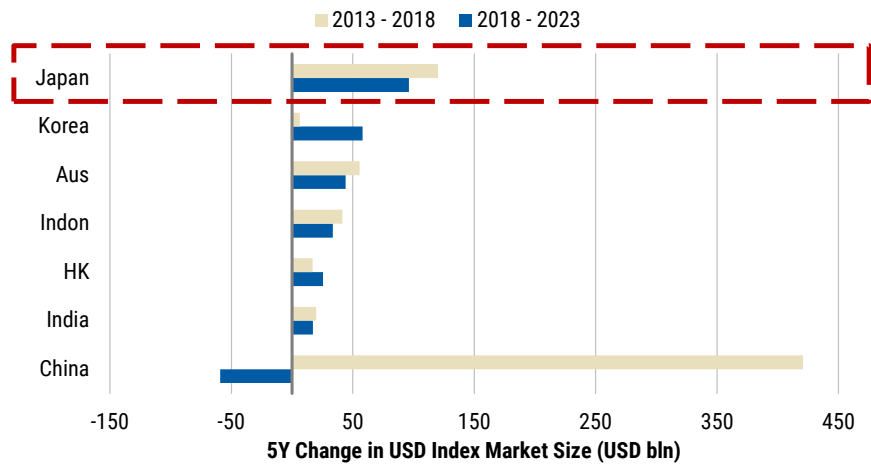
Exhibit 25: ...due to China credit's large negative net supply



Source: iBoxx, Morgan Stanley Research

Given China credit's rapidly shrinking market size, Japan credit's strong positive net supply in the past few years has been a crucial source for Asia credit investors to add credit risk within the region. Between 2018-and 2023, Japan credit's market size grew by US\$120 billion, while China credit's market size shrank by US\$60 billion (Exhibit 26). In hindsight, China credit's rapid growth of US\$421 billion between 2013 and 2018 was not sustainable, while Japan credit's growth in the past decade has been more sustainable. In the next 12 months, we continue to expect Japan credit to have positive net supply and remain a key source of credit risk for Asia credit investors.

Exhibit 26: Japan credit's steady market size growth is overshadowed by China credit's more volatile growth



Source: iBoxx, Morgan Stanley Research

Fact #3: US Credit Investors Have Exposure to Japan Credit, and it Has Been Increasing

- Global credit investors have exposure to Japan credit through US IG credit indices, and Japan credit is the third-largest country exposure in US IG credit indices (aside from US credits).

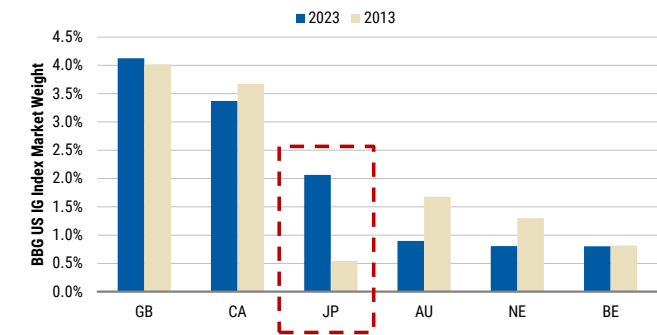
Aside from the US, Japan's country exposure is the third largest and the fastest growing

Besides Asia credit investors, we find that global credit investors also have exposure to Japan credit through the major US IG credit indices. To calculate Japan credit's exposure in the US IG index, we picked three US IG indices that are frequently used by active US IG credit funds and ETFs tracking US IG credit. The three US IG indices we picked are the Bloomberg US IG Index, ICE US IG Index and iBoxx \$ Liquid Investment Grade Index. For example, the iBoxx \$ Liquid Investment Grade Index is the benchmark used by the iShares iBoxx \$ Investment Grade Corp Bond ETF, which currently has an AUM of US\$32.6 billion. Hence, global credit investor support for Japan credit cannot be underestimated.

Based on the three US IG indices we picked, we found that:

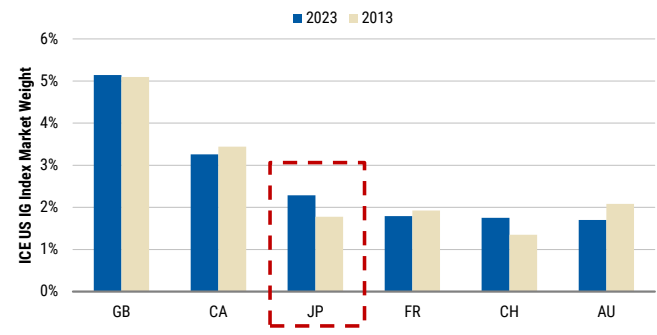
- After US credit, Japan credit is firmly the third-largest country exposure for all three US IG indices ([Exhibit 27](#), [Exhibit 28](#) and [Exhibit 29](#)).
- In the past 10 years, aside from US credit, Japan credit's market share in all three US credit IG indices has the highest increase among countries in the indices.
- Japan credit exposure was in the bottom three country exposure in 2013 for the iBoxx \$ Liquid IG index, the lowest among the three US IG credit indices.
- Japan credit saw the largest market share increase in the iBoxx \$ Liquid IG index ([Exhibit 30](#)) due to the growth of Japan credit's market size that we flagged in [Japan's LC corp bond market post-Covid growth is the highest since 2000](#). Japan credit's market size growth is important because an issuer would need to have a minimum size of US\$2 billion bonds outstanding in order for it to be included in the iBoxx \$ Liquid IG index.

Exhibit 27: Within the three benchmark US IG indices...



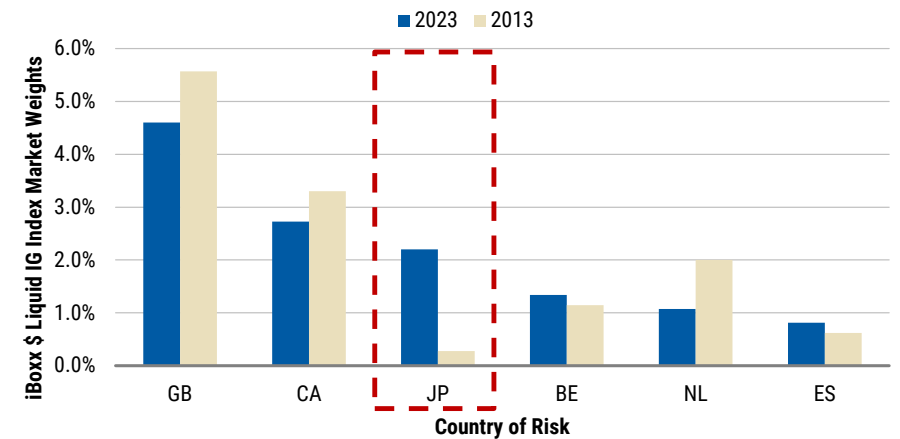
Source: Bloomberg, Morgan Stanley Research

Exhibit 28: ...Japan is the third-largest exposure...



Source: ICE, Bloomberg, Morgan Stanley Research

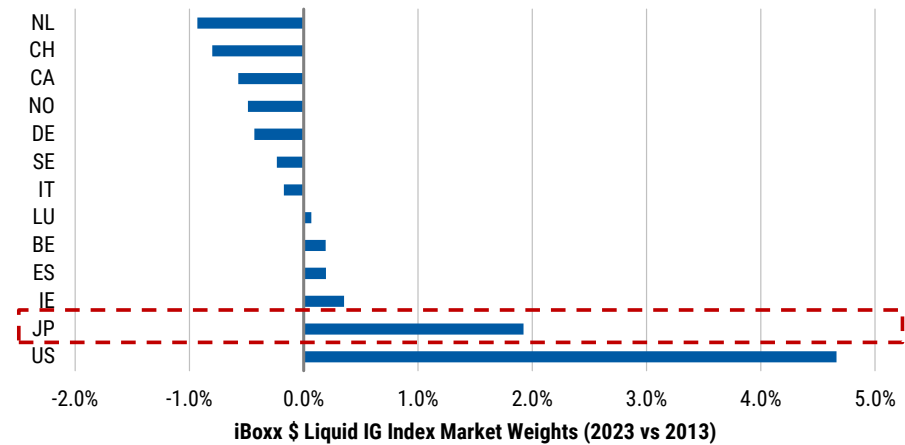
Exhibit 29:
...aside from US credits



Source: iBoxx, Morgan Stanley Research

We acknowledge that Japan credit's current exposure in US IG credit indices remains low at 2-3%. However, if Japan credit continues to grow at the current pace, we would expect global credit investors to start focusing more on Japan credit.

Exhibit 30: Japan credit's market share in the iBoxx \$ Liquid IG index saw the largest increase, outside of US credit



Source: iBoxx, Morgan Stanley Research

Fact #4: Japan USD Credit Lacks Sector Diversification Currently, but This Could Change

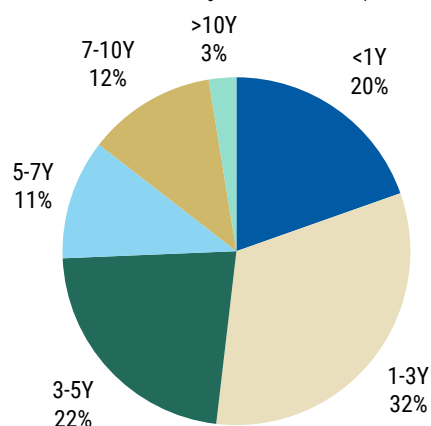
- Japan USD credit could benefit Asia credit investors who need more intermediate duration risk or US credit investors who need more front-end exposure.
- Asia and US credit investors who lack single A rated credit exposure could gain exposure through Japan USD credit.
- Japan USD credit currently lacks sector diversification, but we think that this will change, as we expect higher Japan non-financial USD corp bond net supply.

Japan USD credit's maturity is skewed towards intermediate duration

When we look at Japan USD credit's maturity buckets ([Exhibit 31](#)), we find that slightly half of its maturity is less than 3Y. This high concentration in short maturity is due to the financial sector comprising 60% of the Japan USD credit's sector breakdown (see [Japan USD credit lacks sector diversification](#)). Unlike non-financial corporates, financials (especially banks) tend to prefer to issue in the front end. If we look at the intermediate-maturity bucket (3-10Y), it makes up 45% of Japan USD credit. The long end (>10Y) has the lowest-maturity bucket at only 3%.

Exhibit 31: 45% of Japan USD credit has maturity in the belly (5-10 years)

Japan USD Credit Maturity Breakdown (USD 334 bln)

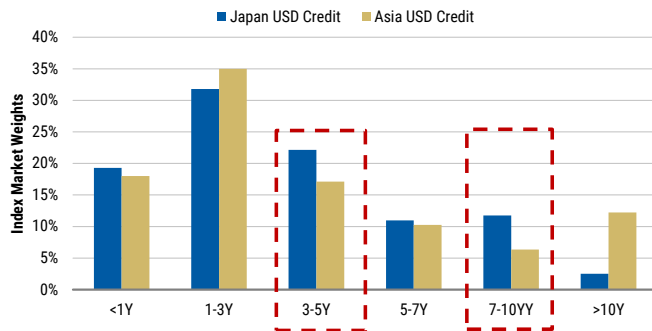


Source: iBoxx, Morgan Stanley Research

We believe that Japan USD credit provides an opportunity for Asia credit investors to add intermediate maturity as its 3-5Y bucket and 7-10Y bucket skews are much higher than for Asia USD credit ([Exhibit 32](#)). In contrast, Japan USD credit provides a good source of

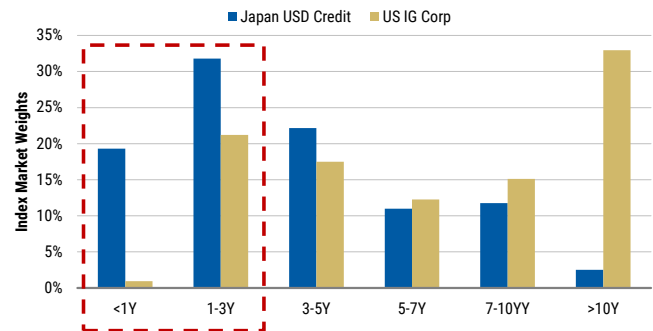
front-end maturity exposure for US credit investors, given that its skew in the front end is significantly higher than that of US IG corps ([Exhibit 33](#)).

Exhibit 32: Compared to Asia USD credit, Japan USD credit's maturity is skewed towards the belly



Source: iBoxx, Bloomberg, Morgan Stanley Research

Exhibit 33: Japan USD credit's maturity is very skewed towards the front end compared to US IG corp



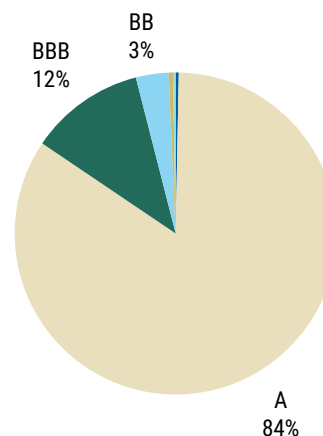
Source: iBoxx, Bloomberg, Morgan Stanley Research

Japan USD credit is a good source of single A credit risk

Japan USD credit's rating distribution lacks any skew because 84% of its bonds are rated single A ([Exhibit 34](#)). The Japan sovereign foreign currency credit rating is between A+ and A, which tends to put a ceiling on the rest of Japan corporate issuers' foreign credit rating. This explains why the Japan USD credit rating breakdown does not have any credit rating above single A. At the same time, financials make up 82% of Japan USD credit, and the financial sector's credit rating is tied to the Japan sovereign credit rating.

Exhibit 34: Japan USD credit's rating is dominated by single A rated credits

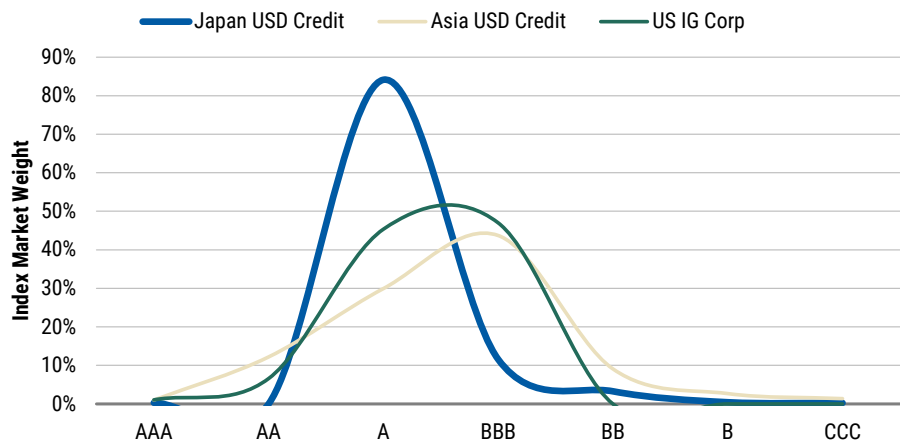
Japan USD Credit Rating Breakdown (USD 334 bln)



Source: iBoxx, Morgan Stanley Research

Not surprisingly, when we compare Japan USD credit's rating distribution with Asia USD credit and US IG corp ([Exhibit 35](#)), it is clear that Japan USD credit has the highest average credit rating among the three. At the same time, we believe that those Asia or US credit investors who want to add single A credit risk should look at Japan USD credit.

Exhibit 35: Japan USD credit is a good source of single A credit exposure for both Asia and US credit investors



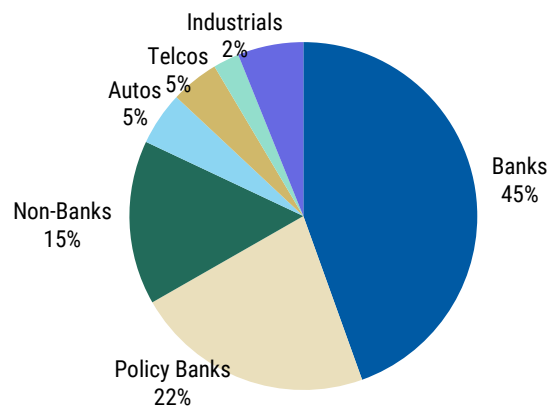
Source: iBoxx, Bloomberg, Morgan Stanley Research

Japan USD credit lacks sector diversification

One main concern we often hear from investors when investing in Japan USD is that it has too much exposure to financials. Financials is currently 82% of Japan USD credit; the top three sectors are all financials – banks, policy banks and non-banks (Exhibit 36). Even after excluding the financial sector, the two largest sectors – auto and telcos – make up close to 60% of the Japan non-financial sector's market share (Exhibit 37).

Exhibit 36: The financials sector is 82% of Japan USD credit sector exposure

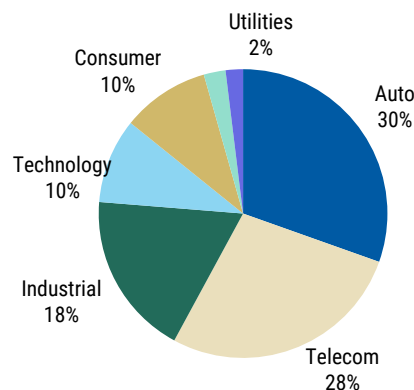
Japan USD Credit Sector Breakdown (USD 334 bln)



Source: iBoxx, Morgan Stanley Research

Exhibit 37: Auto and telecom are currently the two largest sectors in Japan USD credit if we exclude financials

Japan USD Credit Non-Fin POE Corp Sector Breakdown (USD 55 bln)



Source: iBoxx, Morgan Stanley Research

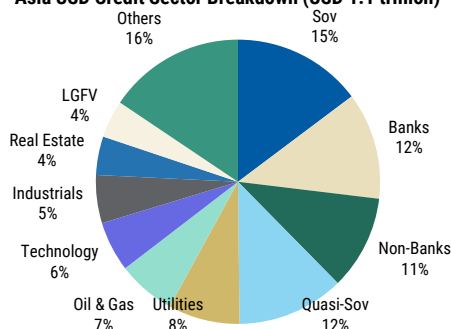
Japan USD credit's high sector concentration risk towards financials is at the extreme end when we compare it to Asia USD credit and US IG corp. Both Asia USD credit ([Exhibit 38](#)) and US IG corp ([Exhibit 39](#)) have only 20-30% exposure to the financial sector. Interestingly, Asia USD credit has only 12% exposure to the banking sector, which means that Japan USD credit could be a source of banking sector risk for Asia credit investors.

That said, we do think that Japan USD credit's high financial sector concentration will decline slowly in the medium-to-long term. We currently see upside potential for Japan USD non-financial net supply, as we expect more Japan overseas M&A deals going forward (see [Overseas M&A is supportive of higher Japan USD corp bond market growth](#)). At the same time, we expect corporate reforms to encourage more Japan non-financial corporates to increase capital efficiency, which should be supportive of higher Japan USD non-financial net supply.

Exhibit 38: The financial sector is only 20-30% sector exposure...

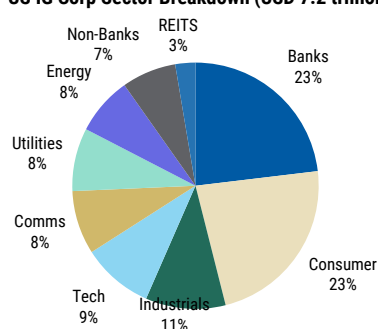
Exhibit 39: ...for both Asia USD credit and US IG corp

Asia USD Credit Sector Breakdown (USD 1.1 trillion)



Source: Bloomberg, Morgan Stanley Research

US IG Corp Sector Breakdown (USD 7.2 trillion)



Source: Bloomberg, Morgan Stanley Research

Fact #5: Japan's LC Corp Bond Market Has Well-Balanced Sector and Duration Distribution

- Japan's LC corp bond market has a balanced maturity profile, in contrast to the maturity profile of most APAC LC corp bond markets, which are skewed towards the front end.
- Japan's LC corp bond market's credit rating distribution looks quite similar to Korea's, but its weighting towards the AAA and AA rating buckets is much lower.
- Japan's LC corp bond market's good sector diversification shows the potential for Japan USD credit to improve its sector diversification.

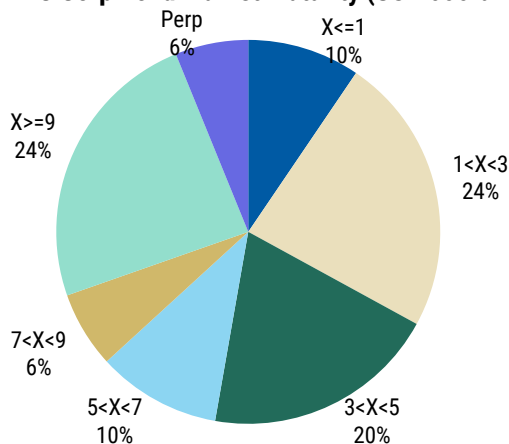
Japan's LC corp bond market has a balanced maturity profile

When we look at Japan's LC corp bond market's maturity breakdown, we find that front-end bonds (less than 3Y maturity) only make up a 34% market share ([Exhibit 40](#)). And when we look at the maturity breakdown of LC corp bond markets in the APAC region, we find that more than half of their corp bond maturities are less than 3Y.

We see two key reasons driving this. First, the LC corp bond issuance tenor tends to be between 3Y and 5Y, and not many APAC corporates can issue LC corp bonds with a tenor above 5Y. Second, the financial sector is usually the largest for most APAC LC corp bond markets, and financial corporates prefer to issue bonds with a tenor between 3Y and 5Y.

Exhibit 40: Close to half of Japan's LC corp bond market maturity is higher than five years

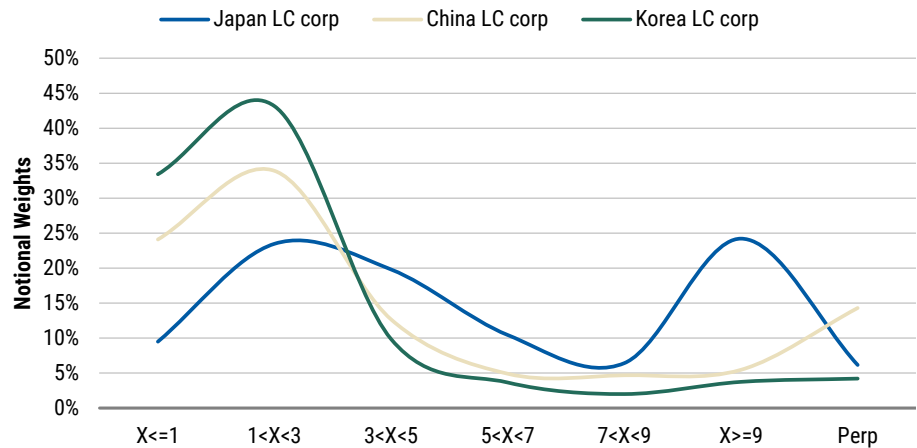
Japan LC Corp Bond Market Maturity (USD 655 bln)



Source: Bloomberg, Morgan Stanley Research

Hence, compared with China and Korea's LC corp bond markets' maturity distribution ([Exhibit 41](#)), Japan's LC corp bond market looks very well balanced. At least 55% of the maturity distribution for both China and Korea's LC corp bond markets is at the front end. More importantly, Japan LC corp bonds stand out by having close to 25% of their maturity being more than 9Y.

Exhibit 41: China and Korea's LC corp bond maturities are highly skewed towards the front end compared to Japan's LC corp bonds

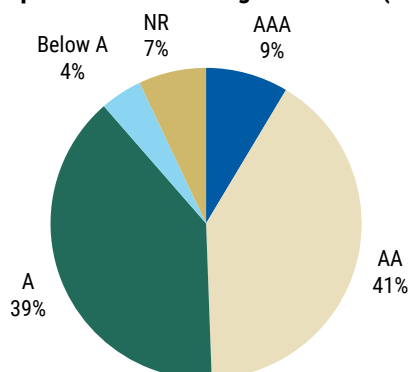


Source: Bloomberg, Morgan Stanley Research

AA and A rated credits make up 80% of Japan's LC corp bonds

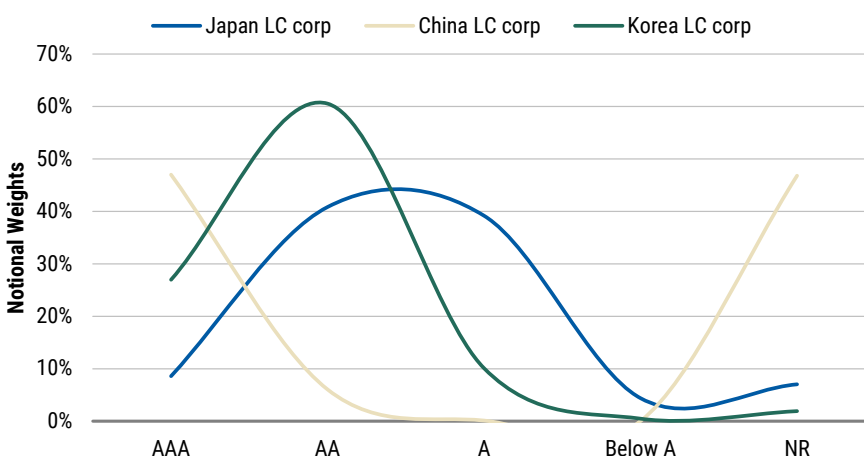
Figuring out Japan's LC corp bond market's credit rating distribution is quite tricky, in our view, because of the difference in Japan's sovereign credit rating between the two local credit rating agencies. Currently, Rating and Investment (R&I) has a AA+ credit rating for Japan sovereigns, while Japan Credit Rating Agency (JCR) has a AAA credit rating. Given that Japan's sovereign credit rating provides a cap on Japan's corporates' credit rating, its LC corp bond market's credit rating distribution varies depending on which credit rating agency is used.

In this report, we use R&I's credit ratings, given that it is the more conservative of the two local credit rating agencies. In addition, it has a larger credit rating coverage of Japan LC corp bonds. For Japan corp bonds that don't have an R&I credit rating, we use JCR's credit rating. Not surprisingly, given R&I's Japan sovereign credit rating of AA+, our Japan LC corp bonds' credit ratings are mainly between AA and single A ([Exhibit 42](#)).

Exhibit 42: AA and single A are the two main credit ratings for Japan's LC corp bonds**Japan LC Corp Bond Market Rating Breakdown (USD 655 bln)**

Source: Bloomberg, Morgan Stanley Research

In contrast, China's LC corp bond market's credit rating follows a rare barbell distribution ([Exhibit 43](#)), where almost half of its ratings are either AAA or not rated (NR). China's LC corp bond market's NR bucket is significantly higher than that of Korea and Japan. Korea's LC corp bond market's credit rating distribution looks quite similar to Japan's LC corp bond market, but its weightings towards the AAA and AA rating buckets are much higher.

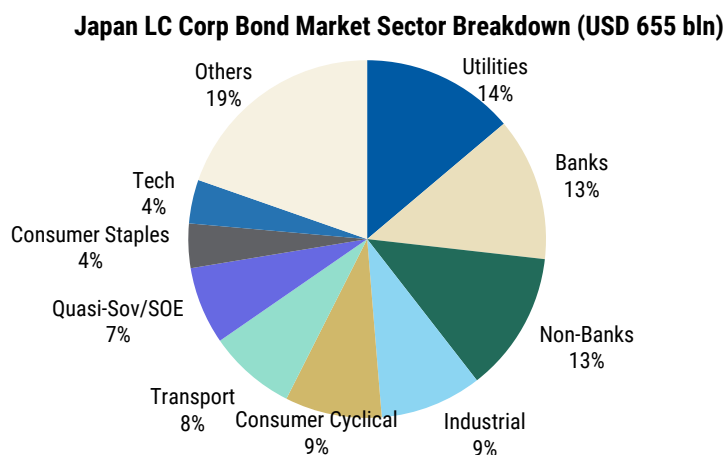
Exhibit 43: Japan's LC corp bond market rating distribution is much closer to Korea's LC corp bond market than China's LC corp bond market

Source: Bloomberg, Morgan Stanley Research

Japan's LC corp bond market has good sector diversification

Unlike Japan's USD credit market (see [Japan USD credit lacks sector diversification](#)), which is heavily skewed towards the financial sector, Japan's LC corp bond market has a very good mix of sectors. The financial sector makes up only 26% of Japan's LC corp bond market versus 82% in the Japan USD credit market. Banks only make up 13% of Japan's LC corp bond sector market share ([Exhibit 44](#)), compared to 67% for Japan USD credit. The largest sector in Japan's LC corp bond market is currently a non-financial sector – utilities.

Exhibit 44: No single sector in Japan's LC corp bond market has more than a 15% market share

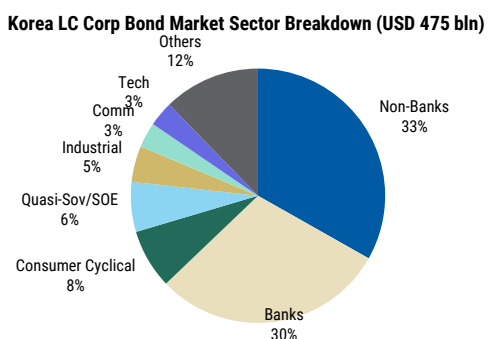


Source: Bloomberg, Morgan Stanley Research

Japan's LC corp bond market has much lower sector concentration risk when compared to Korea's and China's LC corp bond markets. The financial sector dominates Korea's LC corp bond market, with a 63% market share ([Exhibit 45](#)). On the other hand, China's LC corp bond market is dominated by two sectors – LGFV and SOE – that have a combined market share of 77% ([Exhibit 46](#)).

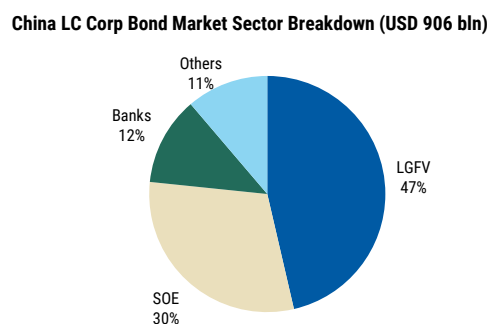
We believe that Japan's LC corp bond market currently shows that Japan's corp bond issuers are from a diversified group of sectors. We think that this could slowly translate to better sector diversification for Japan USD credit. As Japan USD credit continues to grow from here, we believe that its growth could come from the non-financial sector, which would increase sector diversification. We think that we are now approaching the peak of the financial sector's market share for Japan USD credit.

Exhibit 45: Korea's LC corp bond market is dominated by the financial sector...



Source: Bloomberg, Morgan Stanley Research

Exhibit 46: ...while China's LC corp market is dominated by LGFVs and SOEs



Source: Bloomberg, Morgan Stanley Research

Fact #6: Large Japanese Corporates Are Not Averse to Debt Funding

- Large Japanese corporates have been using bank loans, LC corp bonds and FC corp bonds to fund their debt. We expect more Japanese corporates to tap the FC corp bond market as it is a natural progression after being comfortable with tapping the LC corp bond market.
- It is currently cheaper for Japan IG corporates to issue in the USD bond market and hedge it than to issue LC corp bonds, due to tighter Japan IG corporate spreads and a negative JPYUSD cross-currency basis swap.

Corp bonds make up close to 30% of large Japanese corporates' debt funding

One question we often get from investors is how do Japanese corporates fund their debt or do they even have any debt on their balance sheet. One general perception is that Japanese corporates are very cash rich and are averse to using debt.

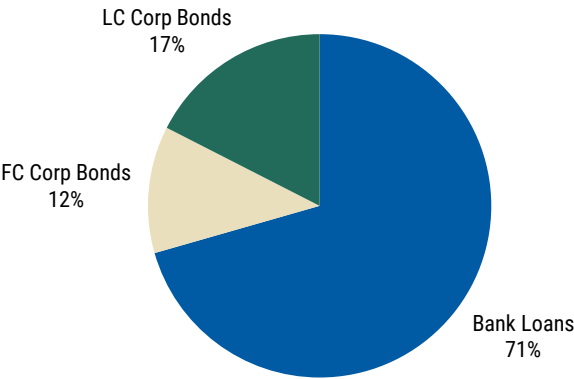
To address this question, we break down 480 large Japanese corporates' debt funding. All 480 Japanese corporates in our sample are non-financial listed corporates that our Morgan Stanley Japan equity research analysts cover. We think that it makes sense to focus solely on listed corporates as they are usually the largest drivers of corporate debt in any country. In addition, listed corporates that have foreign brokers' stock coverage tend to be more exposed to global markets, which means that they are also more likely to issue FC corporate bonds.

We break down Japanese corporate debt into three key sources: 1) Bank loans; 2) LC corp bonds; and 3) FC corp bonds. We estimate that 71% of a large Japanese corporate's debt funding will be in the form of bank loans, 17% will be LC corp bonds and 12% will be FC corp bonds ([Exhibit 47](#)). Within FC corp bond funding, 70% of the funding is in USD. Japan corporates' breakdown is quite similar to the debt breakdown of a typical Asia USD credit issuer, with bank loans being the largest source of debt funding.

We suspect that the current bank loans market share of large Japanese corporates' debt likely peaked as our Japan equity bank team expects Japanese banks' loan growth to slow down in 2024 and 2025 (see [Key Theme 2: Loan Growth Divergence, With DM Decelerating & EM Mostly Accelerating](#), December 13, 2023). We could see corp bonds' market share increase from here to offset the drop in bank loans' market share.

Exhibit 47: Bank loans is currently the largest source of funding for large Japanese corporates

Large Japan Corporates Estimated Debt Funding Breakdown

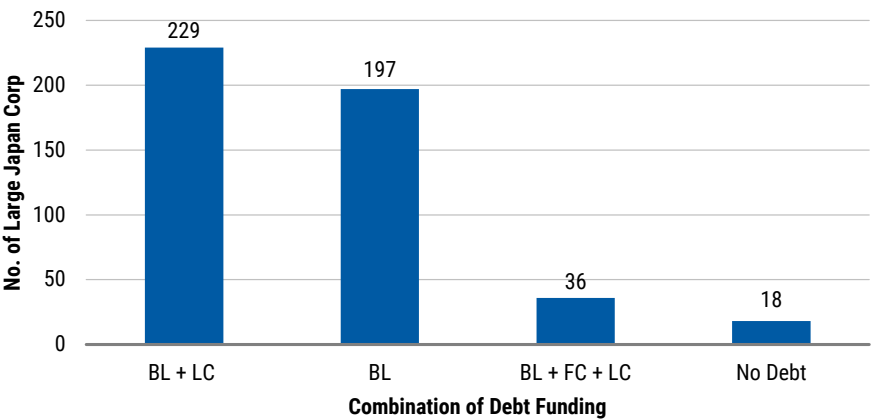


Source: Bloomberg, Morgan Stanley Research

Out of the 480 large Japanese corporates in our sample, only 18 have zero debt on their balance sheet ([Exhibit 48](#)), or less than 5% of our sample. This is in contrast to the general notion that Japanese corporates don't use debt. We find that 41% of Japanese corporates in our sample (197 corporates) have bank loans as their only debt funding source. The rest of the Japanese corporates in our sample have either LC corp bonds or a combination of LC corp bonds and FC corp bonds on top of bank loans as part of their funding.

The penetration rate (7.5% of our sample) of FC corp bond funding currently looks low, and we would argue that there is a lot of room to grow. The good news is that there are 229 Japanese corporates which could issue FC corp bonds, given that they are already tapping the LC corp bond market, which makes tapping the FC corp bond market a natural progression. It is rare to have an APAC corporate that skips tapping the LC corp bond market altogether and moves straight to the FC corp bond market.

Exhibit 48: Close to half of the large Japanese corporates fund their debt through bank loans and LC corp bonds



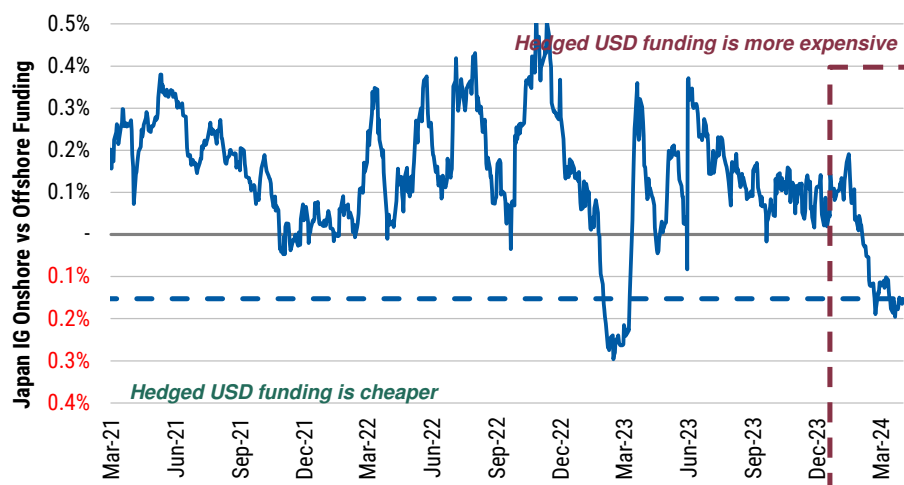
Source: Bloomberg, Morgan Stanley Research; Note: BL = bank loans, FC = FC corp bonds, LC = LC corp bonds.

Hedged USD funding is cheaper than onshore funding for Japan IG corporates

Once a Japanese corporate decides to tap the corp bond market for funding, it has the option to fund in either the FC corp bond market or LC corp bond market. In deciding between FC corp bonds and LC corp bonds, the cost of funding is a key consideration. All things equal, a Japanese corporate should tap the corp bond market that gives the cheapest funding.

However, it is not that simple to compare funding between two different funding currencies given the rates and currency differences. Hence, for FC corp bond funding to be comparable with LC corp bond funding, we would need to hedge the currency and rates risk. For simplicity's sake, we assume that FC corp bond funding is in USD. Hence, a Japan IG corporate could issue a USD bond, hedge its rates and currency risk (using US swap and cross-currency swap), and arrive at hedged USD funding in JPY terms. We can then compare this hedged USD funding with the LC corp bond funding cost that the Japanese corporate would need to pay. The lower of the two would be the cheapest corp bond funding source for the Japanese corporate.

Exhibit 49: It is now cheaper for a Japan IG corporate to issue a USD bond and hedge it than to issue an LC corp bond



Source: iBoxx, Bloomberg, Morgan Stanley Research

Since the start of the year, it has been increasingly cheaper for a Japan IG corporate to issue a USD corp bond and hedge it than to issue a LC corp bond ([Exhibit 49](#)). Currently, if a Japan IG corporate issues a USD corp bond and hedges it, it would save 15bp compared to issuing a LC corp bond. There are two key factors that keep hedged USD funding cheap:

- **Negative JPYUSD cross-currency basis swap:** Although the JPYUSD cross-currency basis swap has turned less negative year to date (due to less demand for onshore investors to hedge their USD investment), it remains negative and at a substantial level of -53bp ([Exhibit 50](#)). A negative JPYUSD cross-currency basis swap is advantageous for Japan IG corporates that want to issue USD corp bonds and hedge them back to JPY.
- **Tighter Japan IG corp credit spreads:** Since the start of the year, the Japan IG corp spread has tightened 22bp to 92bp ([Exhibit 51](#)). It is now just 20bp away from the

historical tight of the index. A low credit spread means cheaper hedged USD funding for Japan IG corporates.

At the same time, LC corp bond funding costs have also increased year to date, driven by higher JGB yields due to the BoJ tightening monetary policy.

Exhibit 50: A negative JPYUSD cross-currency basis swap reduces the USD-hedged cost of funding



Source: Bloomberg, Morgan Stanley Research

Exhibit 51: Tighter Japan IG corp spreads reduce the USD-hedged cost of funding



Source: iBoxx, Morgan Stanley Research

Fact #7: Demand for Japan USD Credit from Investors Has Increased in the Past Two Years

- US credit investors are the largest buyers of Japan USD credit. They are increasing their exposure as a consequence of Japan USD credit's increasing exposure to the US IG corp index.
- Asia credit investors' increased allocation towards Japan USD credit is a function of diversification needs and a need for a new source of credit risk.
- Banks have seen the largest increase in allocation towards Japan USD credit in the past two years due to the lack of high-quality Asia credit net supply, especially in the financial sector.

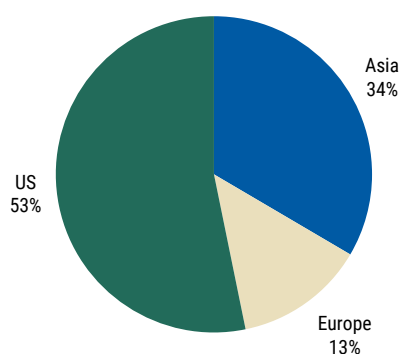
US and Asia credit investors have been increasing their allocation to Japan USD credit since 2022

To get a sense of the buyer base of Japan USD credit, we can look at the primary allocation data for Japan USD bonds. We managed to collect primary allocation data for 127 Japan USD credit bonds that were issued between 2017 to 2023. On average every year between 2017 and 2023, we have primary allocation data for 17 Japan USD bonds. The primary allocation data are mainly divided into two types: geographical and investor.

Based on geographical breakdown data, US credit investors are the largest buyers of Japan USD credit with a market share of slightly more than half ([Exhibit 52](#)). We are not surprised to find that US credit investors are the most important buyer base of Japan USD credit, given that it is currently the third-largest country exposure in US IG credit indices (aside from US credit), which we highlighted in [Aside from the US, Japan's country exposure is the third largest and the fastest growing](#).

Exhibit 52: US and Asia credit investors are two main buyers of Japan USD credit

**Japan USD Credit Geographical Breakdown
2017-2023**

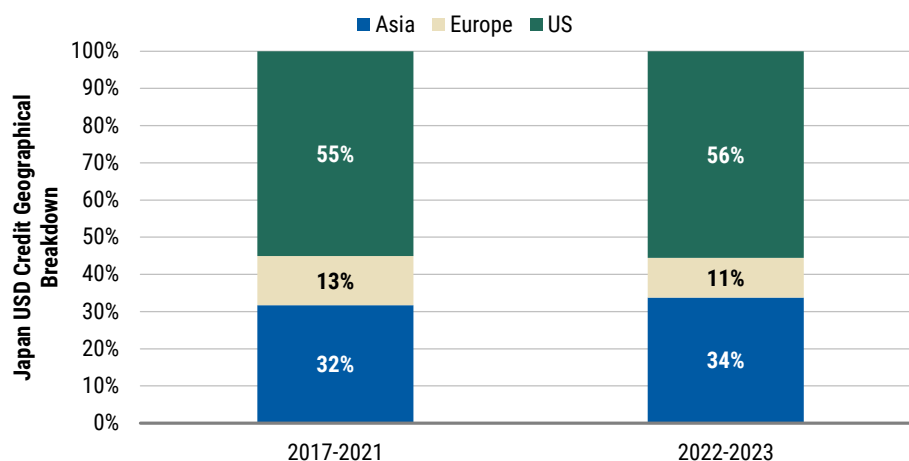


Source: Bloomberg, Morgan Stanley Research

The second-largest geographical buyer base of Japan USD credit are Asia-based investors with a 34% market share, significantly higher than European credit investors. In the last two years, we have seen both US and Asia credit investors increase their allocation to Japan USD credit ([Exhibit 53](#)). Asia credit investors' allocation saw the highest increase to 34% in the past two years, up from 32% between 2017 and 2021.

The drivers of increased allocation towards Japan USD credit for Asia and US credit are quite different. We believe that US credit investors' increased allocation towards Japan USD credit is a function of the increased Japan USD credit exposure in the US IG corp index ([Exhibit 52](#)). On the other hand, Asia credit investors' increased interest in Japan USD credit is a function of diversification needs (see [Japan credit reduces economy concentration risk in APAC credit](#)) and a need for a new source of credit risk (see [Japan credit has been a key source of credit risk for Asia credit investors](#)).

Exhibit 53: Asia and US credit investors have increased their allocation to Japan USD credit in the past two years



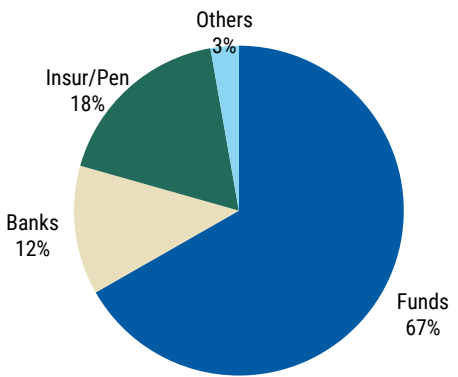
Source: Bloomberg, Morgan Stanley Research

Banks have been incrementally adding more Japan USD credit risk in the past two years

When we look at the investor type breakdown ([Exhibit 54](#)), funds are the single-largest buyers of Japan USD credit, with close to 70% of the allocation. The second-largest investor type are insurance and pension funds, with an 18% market share. We suspect that insurance and pension funds' strong interest in Japan USD credit is due to their sizeable (15% of Japan USD credit) long-end bonds in Japan USD credit ([Exhibit 31](#)). Finally, the third-largest buyers of Japan USD credit are banks, with a 12% market share.

Exhibit 54: Funds have been the main buyer base of Japan USD credit among our three main investor types

**Japan USD Credit Investor Type Breakdown
2017-2023**

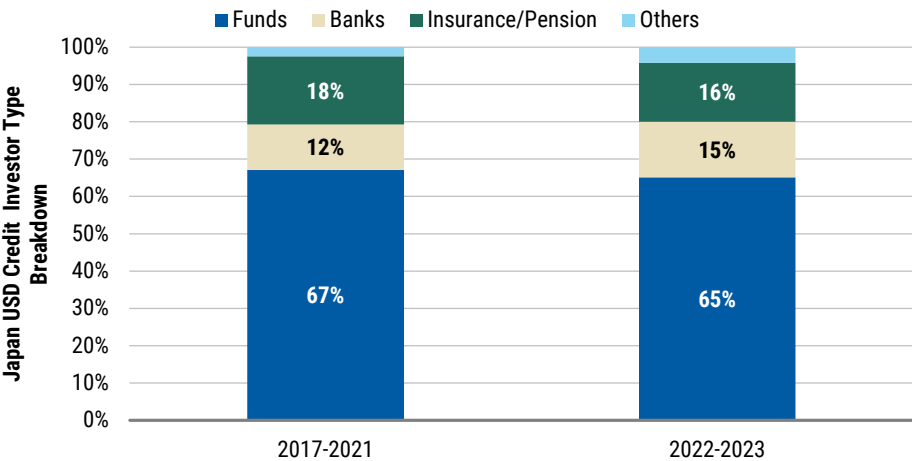


Source: Bloomberg, Morgan Stanley Research

In the past two years, banks have seen the largest increase in allocation towards Japan USD credit among investor types ([Exhibit 55](#)). We believe that the increase in banks' allocation is due to the negative net supply in high-quality Asia USD credit in the past two years, which has driven Asian banks to look at new sources of credit risk within the region. This is especially true for Chinese banks, where China USD credit saw the largest drop in net supply within Asia credit in the past two years.

Lastly, we think that banks, especially Asian banks, are quite comfortable taking financial credit risk, given their existing banking relationships. The Japan USD credit market has a high sector concentration risk towards financials. Hence, for some Asian banks it might be natural to increase allocation towards Japan USD credit, especially given the lack of supply in the Asia financial sector in the past few years.

Exhibit 55: Japan USD credit has seen an increased allocation from banks in the past two years



Source: Bloomberg, Morgan Stanley Research

Fact #8: The Japan IG Spread Looks Cheap versus Global Peers, Especially on a Spread/Duration Basis

- Despite its tight spread, we see Japan IG having the most room to tighten among global IG peers because it is trading at the 16th percentile of its historical spread.
- Japan IG has the shortest duration among global IG peers and looks cheap versus its global peers on a spread/duration basis, thanks to the global IG long-end spread being rich.

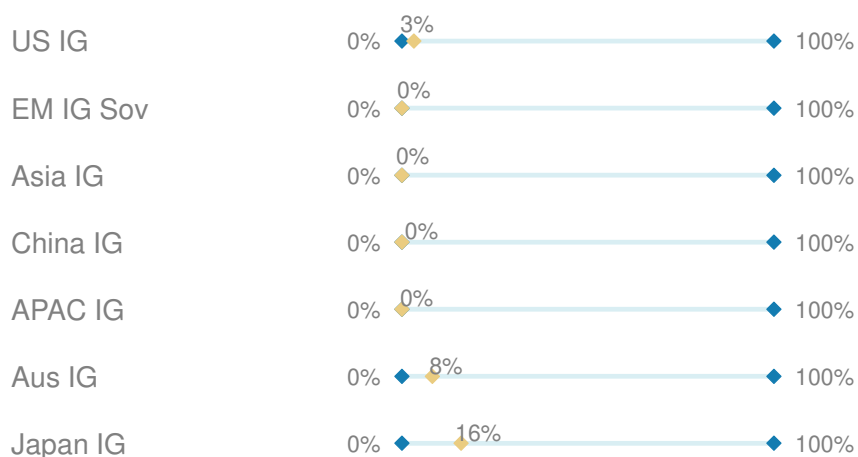
The Japan IG credit spread has the most room to tighten among global IG peers

Given how fast global IG credit spreads have tightened so far this year, we are often asked by clients which asset class has room to tighten more from here. One way to answer this question is to look at the spread history of each global IG credit class and see how far it is from the tights, i.e., percentile rank based on the spread history. For percentile rank calculation, our spread history starts more than 10 years ago, in January 2013.

Out of the eight USD IG credit asset classes we picked, Japan IG really stands out as the one with the most room for spread tightening as its current spread is still a distance from its historical tights ([Exhibit 56](#)). In contrast, Asia IG, China IG and APAC IG are now trading at their historical tights. Although US IG, EM IG sov and Aus IG are not at their historical tights, the room for spreads to tighten further is pretty limited.

Exhibit 56: From a spread percentile rank, Japan IG is currently the cheapest global IG credit

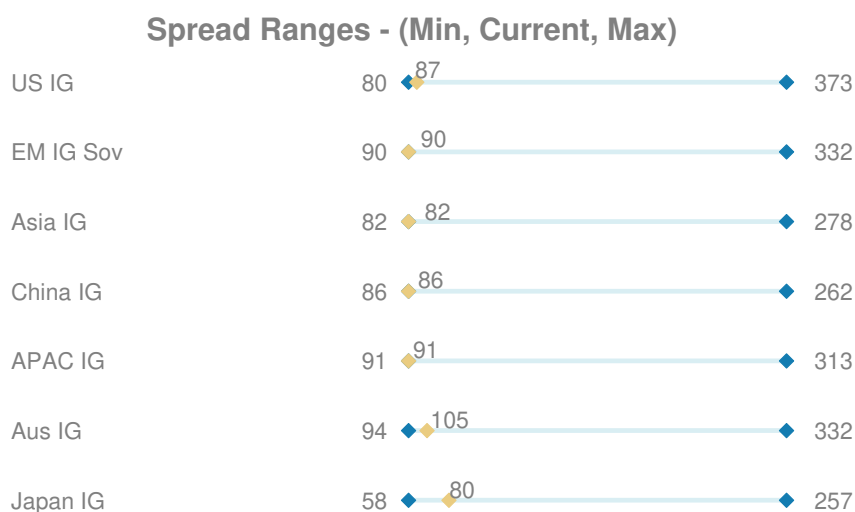
Spread Ranges - Percentile Rank (Current vs. History)



Source: iBoxx, Bloomberg, Morgan Stanley Research; Note: All IG credit history starts from January 1, 2013.

That said, we need to point out that the Japan IG spread is actually the tightest among global IG credits ([Exhibit 57](#)). However, it has a much tighter historical spread, which explains why its current spread is ranked in the 16th percentile. In addition, we believe that the Japan IG spread, structurally, should have a lower spread when compared with global IG credits as it has the shortest duration and highest credit rating among them. Hence, the fact that the Japan IG spread is now the tightest among global IG credit spreads doesn't necessarily limit its room to tighten.

Exhibit 57: Japan IG does not have the widest spread but it has had the tightest spread in the past ten years

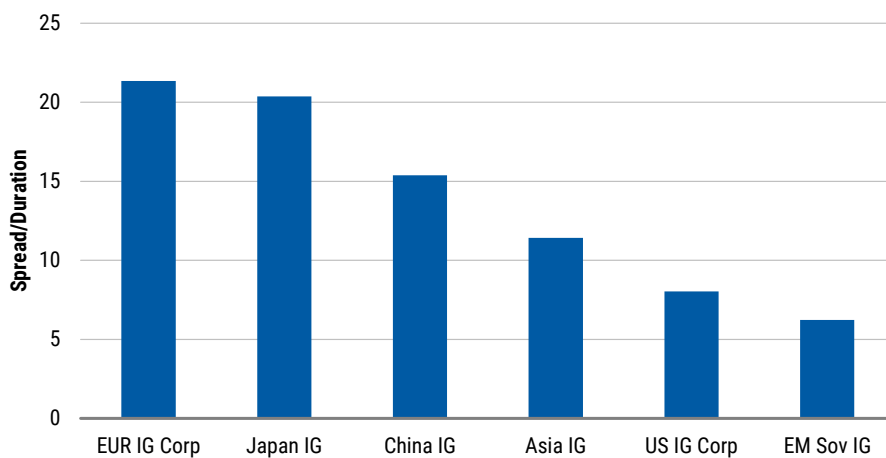


Source: iBoxx, Bloomberg, Morgan Stanley Research; Note: All IG credit history starts from January 1, 2013.

Japan IG credit looks attractive on a spread/duration basis

One way to compare Japan IG credit with global IG credit in a broad sense is to compare spreads with duration (years to maturity). Ideally, the higher the duration, the more the spread compensation. Japan IG credit's duration of 3.9 years is the lowest among all global IG credit indices, which somewhat justifies it having the lowest spread. In contrast, EM sov IG has the longest duration of 14.5 years and US IG corp has the second longest at 10.8 years, but neither of these IG credits have the widest spread.

If we look at the spread/duration for each of the global IG credits, Japan IG credit looks attractive from a duration-adjusted basis. Its current spread/duration comes in a close second to EUR IG corp ([Exhibit 58](#)). The two IG credits with the longest duration among global IG credit end up having the lowest spread/duration. As highlighted by our US credit strategy team, US IG corp long-end spreads are currently very rich compared with their historical valuation (see [Curve forecast – It's all in the 'twist'](#), March 12, 2024), which explains US IG corp's currently low spread/duration number.

Exhibit 58: Japan IG has the second-highest spread/duration among global IG credit

Source: Bloomberg, Morgan Stanley Research

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1458	38%	319	44%	22%	639	40%
Equal-weight/Hold	1730	45%	328	46%	19%	722	45%
Not-Rated/Hold	2	0%	0	0%	0%	1	0%
Underweight/Sell	625	16%	70	10%	11%	232	15%
Total	3,815		717			1594	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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